



Fig. 6. LLM-based agents' cyberattack capabilities of Zero-day attacks.

cloud networks while proposing safeguards against hallucination and bias. Transitioning from defence to offence, Fang *et al.* [61] demonstrate that GPT-4 armed with publicly available CVE descriptions automatically reproduces 87% one-day exploits with performance dropping to 7% absent that auxiliary knowledge, thereby exposing both the promise and the limits of current models. Extending to the mobile domain, Feng *et al.* [64] present AdbGPT, which reproduces 81.3% of 88 Android bugs within 253.6 seconds and surpasses 90% accuracy in step-to-reproduce extraction through prompt engineering and CoT reasoning. Situating these advances within the broader tooling landscape, Ferrag *et al.* [66] critique the pattern dependence of traditional scanners and argue that deep-learning pipelines must reconcile formal-verification precision with real-time performance to scale beyond curated datasets. Building on LLM-assisted exploitation research, subsequent work focuses on vulnerability detection, progressing from code finetuning through knowledge retrieval to domain adaptation. Shestov *et al.* [174] finetune WizardCoder for Java vulnerability detection, framing the task as question-answering and mitigating a 20× class skew with curriculum learning, active sampling, focal loss, and sample weighting. Evaluated on 624 vulnerabilities from 205 OSS projects, their model surpasses CodeBERT-like baselines in both ROC-AUC and F1 on balanced and imbalanced test sets, albeit with noted sensitivity to noisy labels.

Benchmarks: Du *et al.* [57] introduce Vul-RAG, which constructs a knowledge base from 2,174 CVEs and matches candidate functions by semantic retrieval before prompting GPT-4 to reason about causes and fixes. On the new PairVul benchmark, which consists of 4,667 function pairs, Vul-RAG lifts overall accuracy by 12.96% and pairwise accuracy by 110% over prior art while acknowledging leakage risks and its Linux-kernel focus. Ghosh *et al.* [73] target the medical-device supply chain with CVE-LLM, combining domain-adaptive pre-training on regulatory notifications with a human-in-the-loop workflow. Over a two-month pilot, the system materially reduces analyst effort, though long-sequence handling and spurious text in Llama-2 variants remain open challenges.

3.2.3 Honeypot Deployment. Honeypots are controlled environments that mimic vulnerable systems to study adversarial behavior safely. LLM-based agents are deceptive frameworks that generate realistic system responses to attacker inputs. Through contextual analysis and response generation, these agents can simulate authentic behaviors from Linux shells to industrial protocols. Reti *et al.* [158] launch this trajectory by testing 210 prompt templates across GPT-3.5, GPT-4, Llama-2, and Gemini on 1.6 M leaked ClixSense credentials, showing that LLMs can craft honeywords and robots.txt tokens with a 56% indistinguishability rate. Building on automation, Fan *et al.* [60] introduce HoneyLLM, a Go-based medium-interaction honeypot whose GPT-4-Turbo, Claude 3 Opus, and Gemini 1.5 Pro back-ends successfully execute 21 of 25 Linux commands and log both network and system-level activity, as quantified by their ShellEval metric, while highlighting open challenges in latency and jailbreak resistance. Complementing proprietary models, Otal *et al.*

Table 5. Benchmarks for LLM-based cyberattack agents: Main advantages & limitations

Benchmark	Task Focus	Main Advantages	Key Limitations
Safety / Red-Teaming			
AgentHarm [22]	Harmful-instruction	Fully automated evaluation	Text-only prompts
HarmBench [126]	Unsafe behavior robustness	Per-class breakdowns	Focuses only on single-turn prompts
R-Judge [210]	Safety-risk awareness	Multi-step safety scoring	Small scale
Knowledge Q&A / Retrieval			
CS-Eval [209]	Cybersecurity Q&A	Separates knowledge vs reasoning	No interaction or action execution
SecQA [121]	Multiple-choice queries	Simple and fast diagnostic	Small MCQ set; lacks deep reasoning
CmdCaliper [92]	Command safety	Retrieval-based	Synthetic queries
PRIMUS [208]	Corpus assessment	Large-scale domain corpus	No downstream task linkage
CTIBench [14]	Threat intel from CTI reports	APT/malware alignment tasks	Expensive labeling
Pen-Testing / Exploitation			
CyberSecEval 1 [33]	ATT&CK tactics	Safe sandbox testing	No end-to-end chaining
CyberSecEval 2 [32]	Prompt injection	Targets specific exploit types	Limited kill-chain scope; static
AutoPT-Sim [192]	In simulated networks	FSM planning improves ASR	Shell error rates persist
AutoPenBench [74]	Containerized pen-test tasks	Diverse exploit goals	Requires expert setup
Breach-Seek [19]	Multi-agent coordination	Demonstrates role-based planning	Evaluation unclear
Vulnerability & Code Analysis			
Vul-RAG [57]	Function-level matching	Boosts patch accuracy	Limited to known CVEs
PairVul [57]	Code pair vulnerability	Strong pairwise matching	Potential overfitting; dataset-specific
RedCodeAgent [79]	Unsafe code generation	72% attack success	Shell-centric; lacks broader context
Social Engineering / Phishing			
PEN [42]	Phishing mail generation	Human realism evaluations	Only text; small scale
RE-OmniGuard [110]	Multi-turn SE detection	Persona-aware detection	Early-stage sim; unreleased dataset
HoneyPot / Shell Evaluation			
ShellEval [60]	Shell realism and deception	Command match rate	Linux-only; limited in size
LLMPot [187]	ICS honeypot interaction	Byte-level metrics on protocol	Limited function length
Capture-the-Flag (CTF)			
HackSynth [131]	Autonomous CTF solving	Solves 34% of tasks	Lower performance on complex logic
InterCode-CTF [200]	Interactive CTF coding tasks	ReAct&Plan boosts solve rate to 95%	Gaps in binary/reversing domains

[139] fine-tune Llama-3-8B on 617 attacker commands, achieving cosine and Jaro-Winkler similarities of 0.695 and 0.599 to ground-truth outputs, yet note fingerprinting vulnerabilities and the need for adaptive rate-limiting. Human-subject validation follows in Sladič *et al.* [176], where shellLM (GPT-3.5-turbo-16k) deceives participants in 90% of 226 SSH-shell interactions, despite occasional hallucinations and response lag. Domain-specific generalisation is tackled by Vasilatos *et al.* [187], whose LLMPot emulates industrial-control protocols via GPT-4, Llama, and ByT5, attaining 93% Response-Validity Accuracy and 88% Byte-to-byte Comparison Accuracy, though unbounded-length functions remain problematic. Finally, Volkov *et al.* [188] demonstrate real-world viability over a three-month public deployment. Their LLM-augmented SSH honeypot records 8.13 M interaction attempts, which identifies eight autonomous prompt-injection attacks, and observes that LLM-based agents reply within a 1.7s median, which is far faster than humans.

Table 6. Comparison of representative LLM-Enabled cyberattack methods on static-infrastructure networks.

Ref.	Agent Architecture	Network Type	Attack Goal	Blue-team Impact
[175]	ReAct planner & multi-tool orchestration	6G Core & RAN	One-shot break, long-term persistence	Defences largely unaffected (legacy rules bypassed)
[84]	Role-split multi-agent (scan/exploit/privilege)	Enterprise Networks	Privilege escalation and lateral movement	Existing identity and segmentation measures can be bypassed
[147]	Log RAG & anomaly reasoning loops	Data Center Networks	Zero-day detection or abuse of control plane APIs	Alert fatigue decreased; detection improved
[180]	Tokenized flow-based classification with BERT	Software Define Networking	Flow rule manipulation, stealth DDoS	Signature-based IDSs evaded; new attack paths open
[93, 211]	Prompt completion & ICS payload synthesis	Smart Grid	False-data injection, phishing, system spoofing	Real-time model outputs bypass legacy sensors
[9]	Code generation & classical/quantum planning	Quantum Networks	Side-channel attacks on QKD, device layer threats	Control-plane defenses need upgrade

3.2.4 Capture the Flag Challenges. The results that LLM-based agents are being tested on Capture-the-Flag (CTF) challenges mean we can observe their problem-solving strengths and weaknesses. Early evaluations of LLMs for cybersecurity education were conducted by Tann *et al.* in [182]. Comparing ChatGPT, Google Bard, and Microsoft Bing on seven CTF exercises and three tiers of Cisco-level factual questions, they reported that ChatGPT solved 6 / 7 challenges and reached 82% accuracy on knowledge items. These results confirmed the pedagogical value of LLMs while exposing two structural weaknesses. Building on these insights, Turtayev *et al.* in [185] reframe CTF solving as an agentic process. Their ReAct&Plan template steers GPT-4o through up to 30 reasoning–action turns, yet most tasks were solved in only 1–2 turns. The approach pushed success on InterCode-CTF to 95%, eclipsing the prior 29% and 72% baselines. **Benchmarks:** To quantify those weaknesses more systematically, Yang *et al.* in [200] created InterCode-CTF, a 100-task PicoCTF-based benchmark. GPT-4 solved 40% of tasks, struggling with complex reverse-engineering and binary-exploitation. Tests with GPT-3.5, Vicuna-13B, and StarChat-16B showed similar limitations. Abramovich *et al.* in [1] developed EnIGMA, enhancing the SWE-agent with new tools and demonstrations. EnIGMA outperforms prior benchmarks but faces challenges in web-exploitation and data leakage protection.

Finally, the overview of cyberattack-oriented benchmarks for LLM-based agents is shown in Table 5.

3.3 Lessons Learned for Blue Teams

- (1) **Frequent Defense Upgrade:** Defensive teams should implement regular updates to security controls and threat intelligence feeds, fix exposed ports, and misconfigurations. Multiple vulnerabilities signal system weakness, especially with automated scanning. AI malware shows distinct markers like machine-written code and unusual API calls. These identifiers help trace origins and assess threat levels.
- (2) **Active Honeypot Deployment for LLM-based Agents:** Blue teams can enhance their defensive capabilities by deploying LLM-augmented honeypots to engage and monitor attackers at scale. These systems serve as valuable intelligence-gathering tools, providing data that helps update detection signatures and defensive playbooks with emerging attack patterns. To maintain effectiveness, teams must focus on keeping honeypots realistic through regular updates to their conversational models and system responses, preventing sophisticated attackers from detecting and evading these defensive measures.

4 Cyberattack Capabilities of LLMs-based Agents on Static-Infrastructure Networks

Static-infrastructure networks are systems with fixed topology and node placement, maintaining stable traffic patterns. LLM-based agents pose cybersecurity threats by automating attacks on static infrastructure networks, including 6G, enterprise, data center, SDN, smart grid, and quantum networks. These agents focus on “one-shot-break, long-term-stay” attacks for persistent attack installation in critical infrastructure. Table 6 summarizes representative LLM-enabled cyberattack methods across static-infrastructure networks, highlighting their architectures, system targets, attack goals, and defensive implications.

4.1 6G Core and Radio Access Networks

LLM-based agents can translate high-level intents into low-level network commands, potentially abusing 6G programmability to alter network behavior maliciously. Beginning with network management, Mani *et al.* [125] establish that state-of-the-art LLMs can translate natural-language directives into valid router, firewall, and orchestration code. Because these same routines can inject flows or deactivate security rules, their work frames LLM-mediated configuration as a dual-use capability. Shifting focus from functionality to vulnerability exposure, Nguyen *et al.* [135] enumerate attack surfaces unique to 6G and argue that LLM autonomy enables real-time, cross-domain exploit generation. Providing quantitative evidence, Singer *et al.* [175] demonstrate with the Incalmo abstraction framework that an LLM-based agent compromised nine of ten multi-host mobile-core testbeds (25–50 hosts each) by chaining reconnaissance, signaling-protocol exploits, and lateral movement. Finally, Andreoni *et al.* [21] and Yigit *et al.* [206] show that the “cost-collapse” of generative AI simultaneously lowers the barrier to sophisticated attacks and overwhelms legacy detection. At the edge layer of the 6G RAN, Rondanini *et al.* [162] propose an LLM-centric malware-detection architecture for resource-constrained edge nodes. The best GPT variant achieves 97% detection accuracy without exporting raw traffic centrally. Zhang *et al.* [213] show in-context learning matches fine-tuning in wireless-network IDSs, with GPT-4 reaching 95% accuracy across DDoS classes. Legashev *et al.* [114] develop a hybrid LLM-LSTM system for wireless backbones, where Gemma-7B achieves 0.89 F1 in malicious classification with 3% error from poisoning.

4.2 Enterprise Networks

In enterprise networks, valuable assets such as public-facing servers and critical internal services are frequent targets of distributed reconnaissance scans, lateral movement, privilege escalation, and distributed denial-of-service (DDoS) attacks [124]. Attackers typically exploit exposed services, misconfigured devices like internal DNS/NTP servers, and unmanaged mobile devices, which can either be direct victims or leveraged as attack amplifiers. The authors in [84] investigate whether LLMs can perform autonomous penetration testing in enterprise networks through a novel prototype designed for Active Directory environments. They develop and evaluate this prototype using two specific models of OpenAI in a realistic simulation environment, demonstrating that LLMs can effectively conduct Assumed Breach simulations by identifying access points and executing lateral movement. In enterprise networks where LLM-based agents can privilege escalation and lateral movement, blue teams should adopt a zero-trust mindset.

4.3 Data Center Networks

Data center networks usually rely on APIs and orchestration. LLM-based agents could exploit these control plane APIs if credentials or misconfigurations are found. For data center networks, Patil *et al.* [147] introduce an LLM system designed to continuously analyze cloud infrastructure logs and telemetry data for potential *zero-day* attack patterns, demonstrating

superior detection capabilities compared to conventional rule-based approaches across multiple historical breach scenarios. Blue teams should strictly enforce least privilege on API keys, rotate them frequently, and monitor API usage patterns.

4.4 Software-Defined Networking

The SDN controller is a high-value target, and LLM-based agents might launch sophisticated DDoS or traffic-manipulation attacks that standard threshold-based systems are not able to catch [7]. Foundational research by AlEroud *et al.* explores the implementation of inference-based intrusion detection systems for software-defined networking controllers [16]. LLM-based agents could reverse-engineer defenses to reprogram flow tables, enabling evasion and link-flooding attacks. Specht *et al.* show SDN architectures can mitigate industrial malware through network path reconfiguration [178]. This research reveals malicious LLMs could exploit southbound API interfaces for worm propagation. The authors in [180] use BERT-base-uncased to transform network flows into natural language for attack detection in SDN. Using the InSDN dataset with normal and attack flows, the system detects DDoS, DOS, Probe, U2R, BFA, and Web attacks through BERT tokenization and Random Forest Classification. The model achieves 99.96% accuracy with 0.9995 precision and recall scores for known and unseen attacks. Defending SDN infrastructures against LLM threats requires understanding their capabilities in reasoning, evasion, flow manipulation, and network telemetry perception. Traditional detection mechanisms risk obsolescence against these autonomous adversaries.

4.5 Smart Grids

Smart grids could face multi-vector attacks orchestrated by AI. In smart grid operations, LLM-based agents might attempt false data injection to mislead grid control systems [117, 138]. Modern simulation platforms such as *GridAttackSim* [112] and *GridAttackAnalyzer* [113] enable researchers to model and evaluate these attack scenarios in controlled environments. The emergence of LLM-based agents has dramatically accelerated this process by automatically generating sophisticated attack graphs for these testbed environments, reducing scenario development time from hours to mere seconds. The reinforcement learning-based detection system developed by Kurt *et al.* [111] exemplifies the evolving complexity of the adversarial landscape, particularly well-suited to advanced generative AI systems. Zaboli *et al.* [211] provide detailed documentation of ChatGPT's capability to generate convincing phishing campaigns using sector-specific terminology and craft targeted Modbus/TCP attack payloads. Ibrahim *et al.* in [93] conduct a comprehensive examination of large model applications in grid cyber-physical systems, highlighting particular concerns regarding prompt-injection attacks targeting control room assistant systems. Li *et al.* [116] present a systematic analysis of LLM-based risks across power generation, transmission infrastructure, and distributed energy resource orchestration systems, while Zhang *et al.* in [217] reveal critical vulnerabilities wherein LLM-generated code can compromise anomaly detection systems within IoT-enabled electrical substations. To defend against emerging threats, monitoring and limiting LLM capabilities in tool orchestration, prompt interpretation, code generation, and adversarial reasoning is crucial. Implementing model alignment, sandboxed execution, and anomaly detection can help prevent LLM-driven cyberattacks in smart grids.

4.6 Quantum Networks

Quantum communications might be theoretically secure in transmission, but the supporting classical infrastructure is still vulnerable to LLM-based agents. By combining pattern-completion, code-generation, and planning skills, LLMs can (i) automate the discovery of implementation-side channels in QKD devices, (ii) craft novel attack graphs that blend classical and quantum layers, and (iii) orchestrate large-scale post-quantum reconnaissance at machine speed. Ajimon and Kumar present the first systematic blueprint in which an LLM is coupled with quantum-protocol libraries to generate